

Kerala PSC Survey Academy

Chain Surveying - Complete Master Notes

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Chapter Overview:

1. Classification: Plane vs Geodetic Surveying
2. Metric Chains: 20m = 100 links, 30m = 150 links
3. Each link = 20 cm = 0.2 metres
4. Gunter Chain: 66 ft, 100 links
5. Revenue Chain: 33 ft, 16 links
6. Engineer Chain: 100 ft, 100 links

Key Principle:

Working from whole to part to prevent error accumulation.

Primary division first, then secondary, then tertiary.

Corrections in Chain Surveying:

Temperature Correction: $C_t = \alpha \times L \times (T_m - T_o)$

Pull Correction: $C_p = (P - P_o) \times L / (A \times E)$

Sag Correction: $C_s = W^2 \times L / (24 \times P^2)$ -- Always NEGATIVE

Slope Correction: $C_h = h^2 / 2L$ -- Always SUBTRACTIVE

Rank 5 Master Formula Sheet

Memorize all standard equations with exact sign conventions

Formula 1: Temperature Correction

$$C_t = \alpha \times L \times (T_m - T_o)$$

Where $\alpha = 11.5 \times 10^{-6}$ per degree C for steel

Formula 2: Pull Correction

$$C_p = (P - P_o) \times L / (A \times E)$$

P = Applied pull, P_o = Standard pull

Formula 3: Sag Correction

$$C_s = W^2 \times L / (24 \times P^2)$$

ALWAYS NEGATIVE -- subtract from measured length

Formula 4: Slope Correction

$$C_h = h^2 / (2 \times L)$$

Always subtractive from measured horizontal distance

Formula 5: True Distance when chain is too long

$$\text{True Distance} = (L' / L) \times \text{Measured Distance}$$

If chain is TOO LONG: True distance > Measured distance

If chain is TOO SHORT: True distance < Measured distance

Solved PSC Numerical Example

Real exam pattern problem with step-by-step solution

Problem:

A 20m metric chain was found to be 10 cm too long after measuring a distance of 1500m. If the chain was accurate before starting work, calculate the true distance.

Rank 5 Step-by-Step Solution:

Step 1: Average error = $(0 + 0.10) / 2 = +0.05$ m

Step 2: Average chain length L-prime = $20 + 0.05 = 20.05$ m

Step 3: True Distance = $(L\text{-prime} / L) \times \text{Measured Distance}$

Step 4: True Distance = $(20.05 / 20) \times 1500$

Answer: True Distance = 1503.75 metres

PSC Exam Trap Warning:

Kerala PSC sets trap options on sign of correction vs error.

Correction = -Error. Chain too long = positive error = negative correction.

Always check: Is chain too long or too short?